

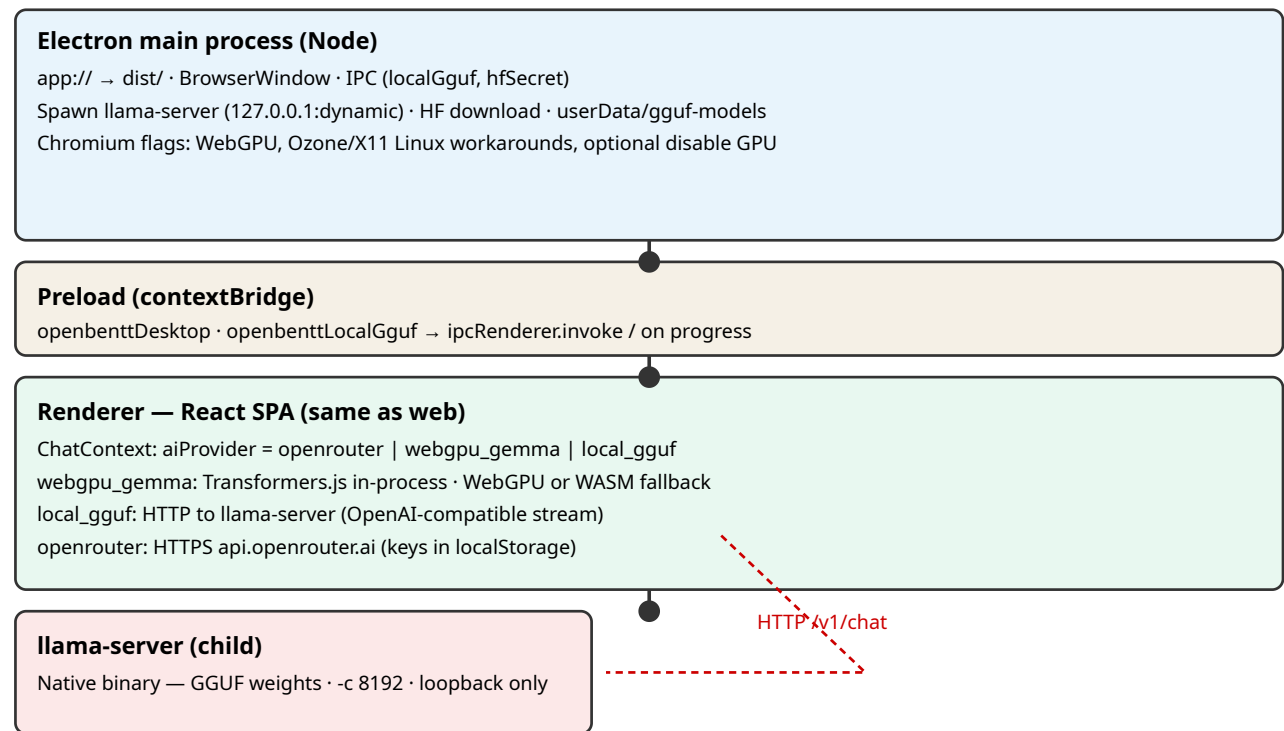
Openbentt — Desktop System Design

Scope: Electron desktop only. Product: Openbentt (local-first chat + OpenRouter + workspaces).

1. Problem solved

- **Privacy / offline-capable AI:** On-device inference via **WebGPU/WASM** (Transformers.js in the renderer) or native **GGUF** via a local **llama-server** child process.
- **One UI everywhere:** Same Vite/React app; Electron adds shell, IPC, and Chromium flags.
- **GGUF without Ollama:** Main process downloads .gguf, manages registry under userData, spawns OpenAI-compatible server on 127.0.0.1.
- **Linux WebGPU:** Early app.commandLine.appendSwitch calls enable WebGPU paths that often fail in stock browser profiles.
- **HF token handling:** safeStorage-backed (or restricted fallback) storage for gated model downloads.

2. System chart (logical)



3. How local models use system parts

Path	Runtime	Data / compute
webgpu_gemma	Inside Chromium renderer	HF repos via Transformers.js cache; GPU WebGPU or WASM CPU; dtype/backend cascade + tiny model fallback.
local_gguf	Separate llama-server process	Registry + files in Electron userData; binary from env / settings / resources/llama/ / PATH; chat code streams like OpenRouter

		but URL = localhost.
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4. Distribution plan

- **Trigger:** Git tag v* → `.github/workflows/release.yml`.
- **Outputs:** Linux AppImage + deb; Windows NSIS exe (+ zip); macOS dmg + zip; optional static web zip.
- **Build:** `npm run build` then `electron-builder` per OS; BusyTeX fetched for LaTeX features.
- **Signing:** Documented as optional; unsigned builds may trigger Gatekeeper / SmartScreen.
- **Note:** Shipping llama-server in installers may require `extraResources` — today users often rely on PATH or manual path.

5. Improvements (desktop)

- Bundled per-platform **llama-server** in releases.
- macOS notarization + Windows code signing.
- Auto-update channel (e.g. electron-updater).
- Strict IPC payload validation; single-instance lock.
- Opt-in crash reporting (no chat content).

6. Key files

```
electron/main.mjs          - window, app:// protocol, GPU/WebGPU flags
electron/preload.cjs       - exposes openbenttLocalGguf to renderer
electron/localGgufService.mjs - registry, HF download, llama-server lifecycle
electron/hfSecretStore.mjs - HF token encryption
src/context/ChatContext.tsx - routes providers by aiProvider
src/lib/gemmaWebGpu/*      - in-browser local LLM
src/lib/localGguf/*        - desktop bridge + GGUF streaming
```

Openbentt desktop system design — print via browser or headless Chromium --print-to-pdf.